Table 6.4 Refractive indices for different wavelengths

Colour	Wavelength in vacuum (nm)	Crown glass	Flint glass
Violet	396.9	1.533	1.663
Blue	486.1	1.523	1.639
Yellow	589.3	1.517	1.627
Red	656.3	1.515	1.622

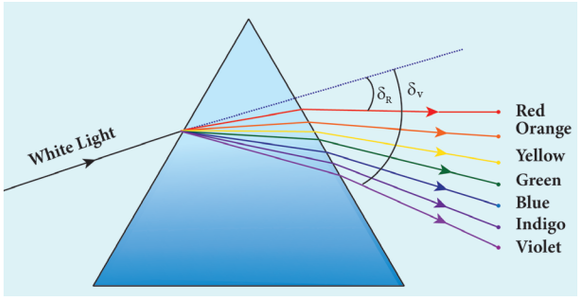


Figure 6.44 Angle of deviation for different colours

6.7.5 Dispersive Power

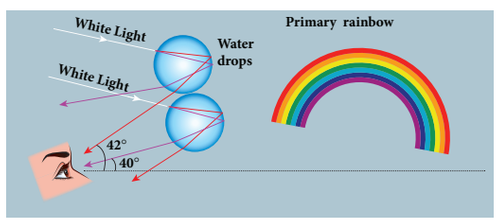
Consider a beam of white light passing through a prism. It gets dispersed into its constituent colours as shown in Figure 6.44.

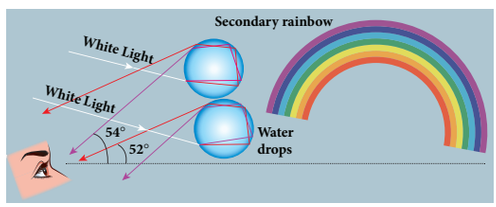
If the angle of prism is small of the order of 10°, the prism is said to be a small angle prism. When rays of light pass through such prisms, the angle of minimum deviation also becomes small. Let A be the angle of a small angle prism and δ be its angle of minimum deviation, then equation (6.89) becomes,



Rainbow appears in sky during mild shower (or) near the fountains/falls where there are water droplets remain suspended in air. A rainbow is seen when the sun is at the back of the observer. Dispersion occurs when sunlight enters a water droplet and the white light is split into its constituent seven colours. A primary rainbow is formed when the light entering a droplet undergoes one total internal reflection inside it. Sometimes, a secondary rainbow is also formed enveloping the primary rainbow as shown in the figure. The secondary rainbow is formed when light entering a raindrop undergoes two total internal reflections. The order of colour in primary rainbow is from violet to red whereas in secondary rainbow it is from red to violet. The angle of view in primary rainbow from violet to red is from 40° to 42°. The angle of view for secondary rainbow from red to violet is from 52° to 54°.







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